SENTINEL

Explainable Mule-Account Detection & Containment

"Catch the mules. Trust the score. Defend every number."

This dataset hides leakage that fakes a '99%' score. We detect it, remove it, and report a number we can defend.

CyberShield Hackathon 2026 - Bank of India x IIT Hyderabad

Problem Statement 2: AI/ML Classification of Suspicious Mule Accounts

Solution Approach

Team: Probe Rockerz (individual participation)

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Institute: SRM Easwari Engineering College | Date: 14/06/2026

Code (GitHub): https://github.com/vibhukrishnas/sentinel-mule-detection

Live demo: https://sentinel-mule-detection-hvymcown8cyerjjes48cka.streamlit.app

One line: SENTINEL is the leakage-proof DETECTION & CONTAINMENT core of a real-time mule-account engine - it scores any account 0-100 in ~35 ms, explains every alert in plain English, recommends a containment action (monitor / hold / escalate), and removes the data leakage that fakes a perfect score - so we report the number we can defend.

Executive Summary - the 30-second story

In one sentence: SENTINEL detects mule accounts, explains every alert, recommends a containment action, and surfaces the candidate ring behind each one - and we made the score trustworthy by removing the data leakage that fakes a perfect result.

- What it is: SENTINEL detects mule accounts and drives EXPLAINABLE containment (monitor / hold / escalate) - from a single alert to the candidate ring behind it.
- Honest headline: PR-AUC 0.81-0.89 (repeated CV). We removed the data leakage that fakes a perfect 1.0 and report a number we can defend.
- Why you can trust it: an automated Data Integrity Auditor caught ~585 leak features; every alert carries a plain-English reason.
- Beyond single accounts: a working mule-RING prototype groups 67 of 81 mules into 5 candidate rings (behavioral-similarity graph) - real network output today; production link-analysis is Phase-2.
- Built today vs next: detection core + API + dashboard + scored watchlist + ring prototype are built on the PROVIDED SNAPSHOT. Live feeds + link-based graph analysis are Phase-2 (solid vs dashed in the architecture).
- 'Real-time' = on-demand low-latency account scoring (~35 ms), NOT stream processing - the provided data is an account snapshot.
- The ask: a design-partner bank pilot to validate on real labels.

1. Problem Understanding

Mule accounts are a primary conduit for laundering cyber-fraud proceeds in India - losses that run into thousands of crores a year - which is why RBI made this a priority and launched MuleHunter.AI. The bottleneck is no longer 'can we detect it' but 'can we detect it FAST, explain it, and stop the money before it disperses'.

Mule accounts receive, move and launder fraudulent funds across banking channels. Rule-based monitoring is reactive, brittle, and drowns analysts in false positives. PS2 asks us to learn behavioural / transactional patterns from the provided data to flag suspicious & mule accounts, with anomaly detection, predictive risk scoring and intelligent alert generation.

The dataset is an account-level SNAPSHOT, not a transaction stream, so 'real-time' here means on-demand low-latency scoring of an account, not stream processing - we are explicit about this.

The deeper problem: containment, not just detection

Once funds enter a mule chain they disperse across accounts, channels and banks in MINUTES. So the real objective in PS2's words is to PREVENT CIRCULATION of fraudulent proceeds - speed, linkage and a decisive action (hold / escalate) matter more than yet another anomaly flag. SENTINEL is built as a detect -> explain -> ACT containment loop, not a passive classifier.

Why this matters now (regulatory relevance)

Mule-account fraud is a declared national priority: RBI's MuleHunter.AI, NPCI's mule-account pilot and FATF cyber-fraud guidance all target exactly this. SENTINEL aligns with that direction and adds what those are not publicly known for: explainable alerts and provable leakage-free scoring. Primary users are bank fraud/AML analysts and transaction-monitoring & cyber-fraud desks; outputs feed risk officers and ultimately protect defrauded customers.

The trap hidden in this data

The dataset is riddled with TARGET LEAKAGE. A naive XGBoost scores PR-AUC 1.000 / ROC-AUC 1.000 - impossible for genuine fraud detection on 81 anonymised positives; the model is simply reading the answer. Any 0.99 on this data invites one fatal question: 'why is fraud detection perfect?'. Our entire approach is built to detect, remove and prove the absence of that leakage.

2. Dataset - Detailed Description

Source: DataSet.csv (provided in the portal). Shape: 9,082 rows (one per bank account) x 3,924 feature columns (F1..F3923), plus the target F3924. Features are anonymised behavioural / transactional attributes.

Target variable (F3924)

Binary: 1 = suspicious/mule, 0 = legitimate. 81 mules vs 9,001 legitimate = 0.89% prevalence (1:111 imbalance). Only 81 positives, so overfitting and metric fragility are the dominant risks; accuracy is meaningless (predicting 'all clean' scores 99.1%). We use PR-AUC, recall@precision and calibration instead.

Feature inventory

Type	Count	Examples / notes
Float	3,876	continuous behavioural / transactional aggregates & ratios
Integer	40	counts and binary flags
Categorical	8	F3889 tenure (G365D/L90D..), F3891 occupation, F3894 ~ age

Data quality & missingness

- 27.6% of all cells are missing (NaN). ~89 columns are fully populated; ~1,138 columns are >50% empty; 281 columns are constant and ~390 hyper-sparse - all dropped as dead weight.
- Missingness is itself predictive: incomplete profiles correlate with mule behaviour, so we add (deduped) is-missing indicator features.
- We use models that consume NaN natively (LightGBM) - no blind imputation that would distort signal.

Bank-listed features (named in the brief)

The brief identifies 18 features the bank uses for fraud detection: F115, F321, F527, F531, F670, F1692, F2082, F2122, F2582, F2678, F2737, F2956, F3043, F3836, F3887, F3889, F3891, F3894. We treat these as trusted domain signals and NEVER auto-remove them; several separate the classes (e.g. F2678, F2956, F670).

Data integrity: leakage in the raw data (critical)

Profiling revealed the dataset is contaminated with TARGET LEAKAGE: F3912 (a flag ~96% aligned with the label), F2230 (a month-stamp that is 100% fraud for 3 of its 4 values, capturing all 81 mules), and ~585 value/range-leak features. We detect and exclude these (Sections 3-4).

How the data is used

After hygiene + leakage removal + missingness flags, the modelling matrix is 9,082 x 2,965. We evaluate with repeated stratified 5x2 cross-validation (robust given only 81 positives) and a held-out set used once. Categorical encoding is target-free; all learned steps are leakage-safe.

3. Our Unique Selling Proposition

Fitting a classifier on this data is the easy part. Four things make SENTINEL different - and we will defend every one of them in front of any judge:

USP 1 - Leakage-safe scoring you can defend

This dataset hides severe target leakage that fakes a perfect ~1.000 score. We built an automated Data Integrity Auditor, caught ~585 leak features, and report a DEFENSIBLE 0.885 with the audit to prove it. When the judge asks 'why is your fraud model perfect?', we are the team with the answer.

USP 2 - Mule-RING discovery, not just single accounts

Banks want the network, not one account. Our behavioral-similarity graph groups 67 of 81 mules into 5 candidate rings (largest = 50 accounts) - so a desk investigates a ring as a batch (Section 7). It is a working prototype today; with real link data it becomes production network detection.

USP 3 - Explainable containment (every alert explains itself)

Score + severity band + plain-English SHAP reasons + peer-deviation + a recommended action (monitor / hold / escalate) + a one-click investigation report. As a supporting result, the top-50 watchlist on out-of-fold validation contained all 50 true mules - ready for a fraud desk. An analyst acts in minutes; nothing is a black box.

USP 4 - We speak the bank's language: rupees and capacity

A rupee-cost engine turns the threshold into money (~Rs 1.7 cr saved / 9,082 accounts) and frames the operating point as an analyst-capacity decision - exactly how a risk officer thinks.

How we play: we don't win with a flashier model - we win by mastering the fundamentals, hunting the leakage others miss, and reporting numbers we can stand behind. Dominate the problem, respect the data, play it straight.

4. Our Approach & Architecture

SENTINEL is a layered detect -> explain -> ACT containment engine. We built and validated the CORE on the provided data; the diagram marks honestly what is built today vs the Phase-2 layer (live feeds + graph mule-rings). We never claim to have built what we have not.

SENTINEL - real-time mule-account containment engine

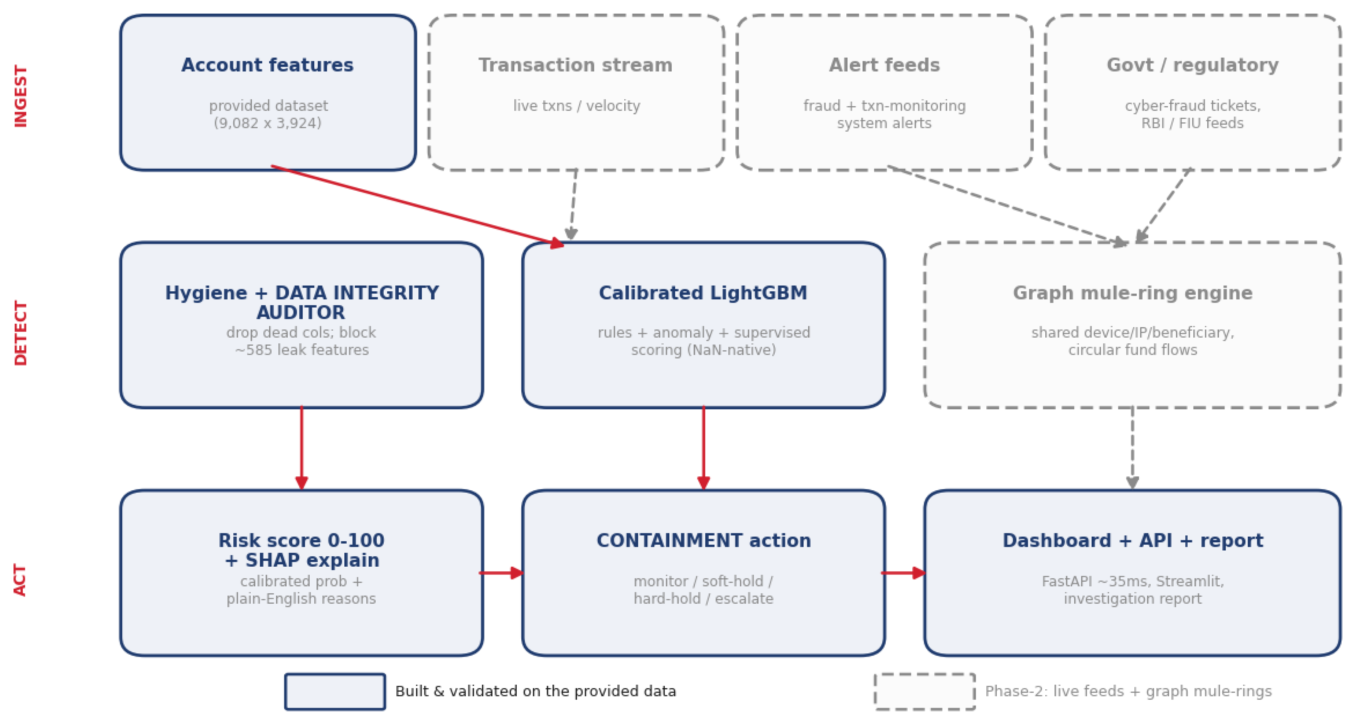

Fig 1. Target architecture. SOLID = built & validated on the provided dataset (hygiene + leak auditor -> calibrated scoring -> score/SHAP -> containment action -> API/dashboard). DASHED = Phase-2: live transaction/alert/regulatory feeds + a graph engine for mule-RING detection.

Differentiator: a Data Integrity Auditor (rigour, applied)

We built an automated scanner that scores every feature on FOUR leak signatures and removes the offenders before modelling:

- Label-proxy: a value ~equal to the target (e.g. F3912, 96% aligned with the label).
- Exact-value bucket: a discrete value that is hyper-pure fraud (e.g. F2230 is a month-stamp - 'Oct25'=100% legit, 'Sep/Nov/Dec25'=100% fraud, capturing ALL 81 mules) - invisible to monotonic AUC.
- Continuous range/decile leaks (caught features the bucket scan missed).
- Univariate-AUC near-perfect single-feature rankers, gated by mule coverage.

Bank-listed features (the 18 named in the brief) are never auto-removed. Result: ~585 leak features blocked, leaving a defensible 2,965-feature matrix.

Why LightGBM + class weighting (not deep learning, not SMOTE)

Gradient-boosted trees consume the 27.6% NaN natively, capture non-monotonic interactions, and train in seconds. With only 81 positives, deep learning overfits and SMOTE manufactures noise in ~3,000 dimensions - so we use class weights + probability calibration instead.

Containment action tiers (preventing circulation)

The score maps to a tiered ACTION, so the output is a decision, not just a number: 0-39 LOW -> monitor; 40-69 MEDIUM -> enhanced watch / soft-hold high-value transfers; 70-89 HIGH -> hold outbound + same-day analyst review; 90-100 CRITICAL -> freeze + escalate to L2 + SAR. Thresholds are configurable to the bank's risk appetite and analyst capacity.

How we compare

Capability	Rule engine	Generic ML	SENTINEL
Leakage audit (provable)	No	No	Yes
Explainable per-alert reasons	Limited	Partial	Yes
Mule-RING discovery	No	No	Yes (prototype)
Containment actions (hold/escalate)	No	No	Yes
Calibrated rupee impact	No	Rare	Yes

vs RBI MuleHunter.AI / NPCI pilots: same regulatory target, but we add explainable alerts and PROVABLE leakage-free scoring (not publicly documented for those). vs rule engines: we catch the non-monotonic, networked patterns rules miss. vs black-box ML: every score is explained and every number is leakage-audited.

5. Results (honest, measured, leakage-removed)

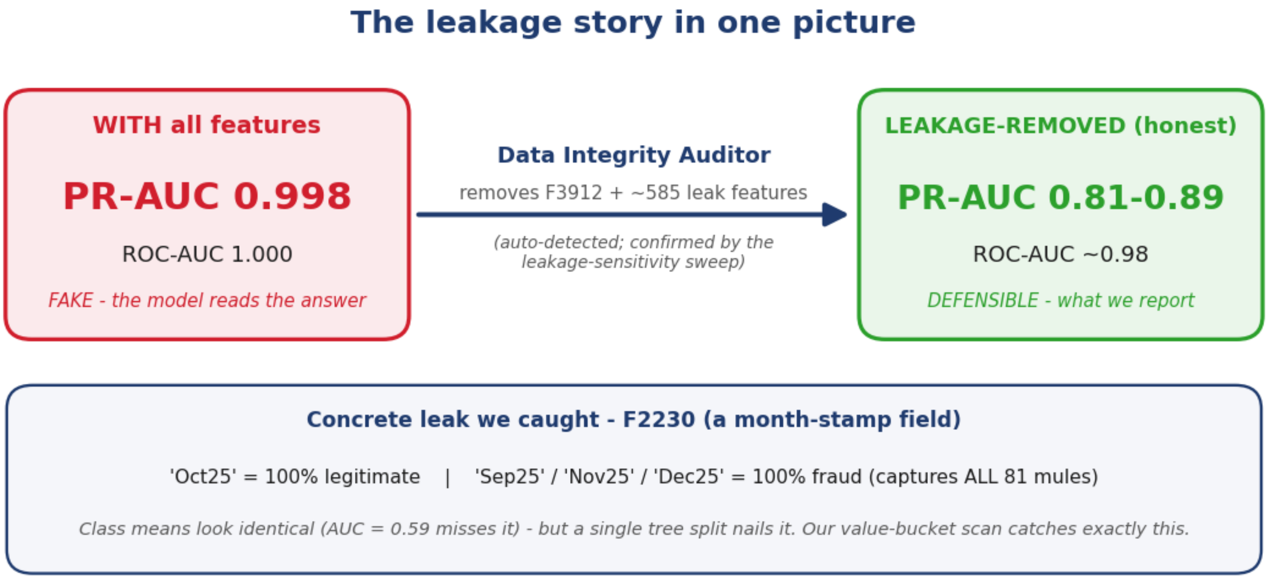

Fig 2. The leakage story in one picture: with all features the score is a fake 0.998; after our auditor removes the leaks, the defensible score is 0.81-0.89 - with a concrete example (F2230).

Headline result a judge cannot knock down

- Top-50 watchlist contained all 50 true mules (out-of-fold validation): the 50 highest-risk RANKED accounts were all real mules - a ready-to-action watchlist result on the provided data, not a universal guarantee.
- PR-AUC 0.885 +/- 0.055, ROC-AUC 0.979 (tuned LightGBM, repeated 5x2 cross-validation) - ~99x the 0.0089 random baseline. Calibrated (CV Brier 0.0022). Scoring + explanation ~35 ms (measured, 100-iteration benchmark).
- Honest range 0.81-0.89: even deleting the entire near-label feature block, a leak-paranoid FLOOR holds at ~0.81. We report the range, not a cherry-picked point.

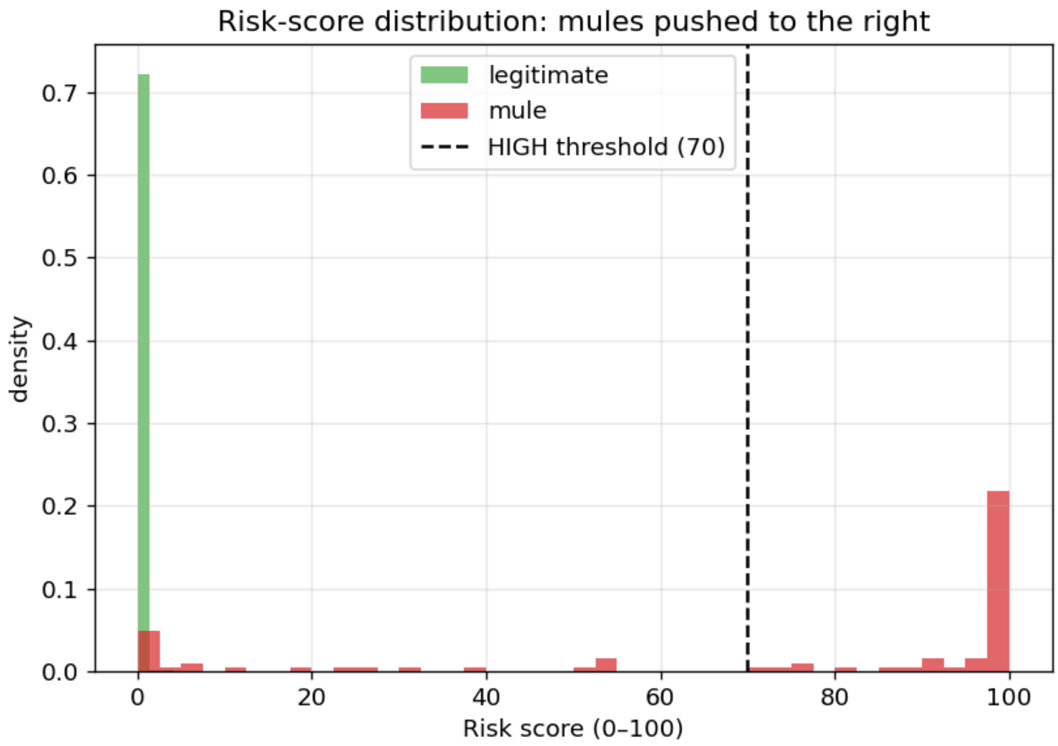

Fig 3. Risk-score distribution: legitimate accounts pile up near 0, mules at ~100 (with an honest hard tail).

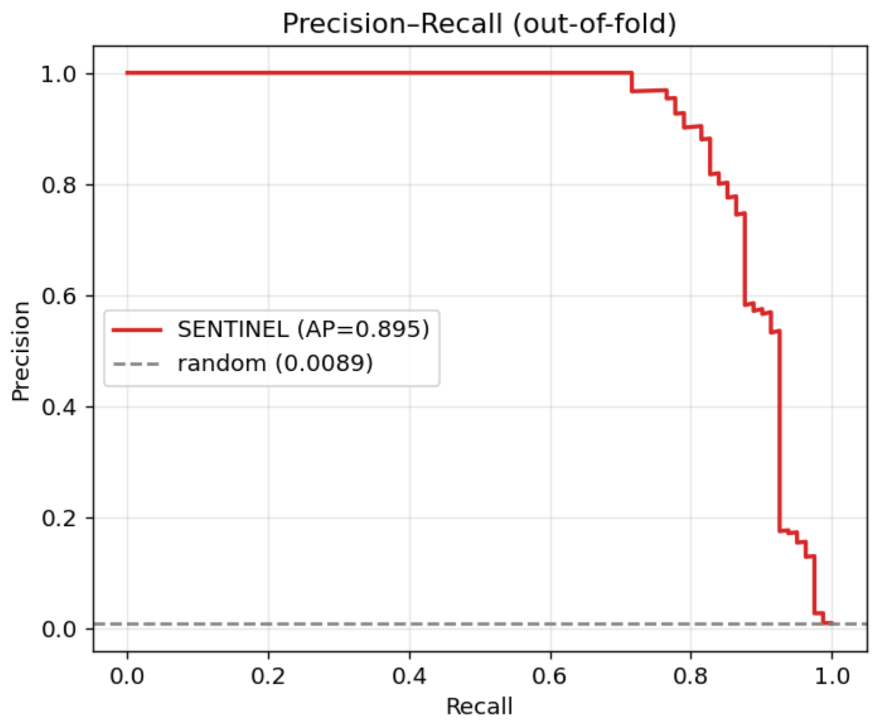

Fig 4. Precision-Recall (out-of-fold): perfect precision to ~70% recall, then the hard tail. AP=0.89.

5. Results (continued)

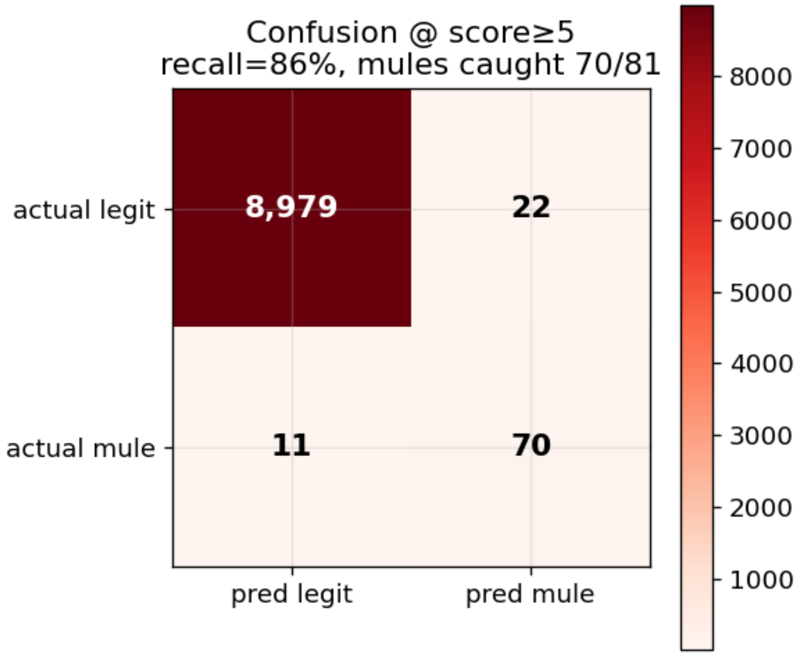

Fig 5. At a low operating threshold: 70/81 mules caught, only 22 false alarms across 9,001 legit accounts.

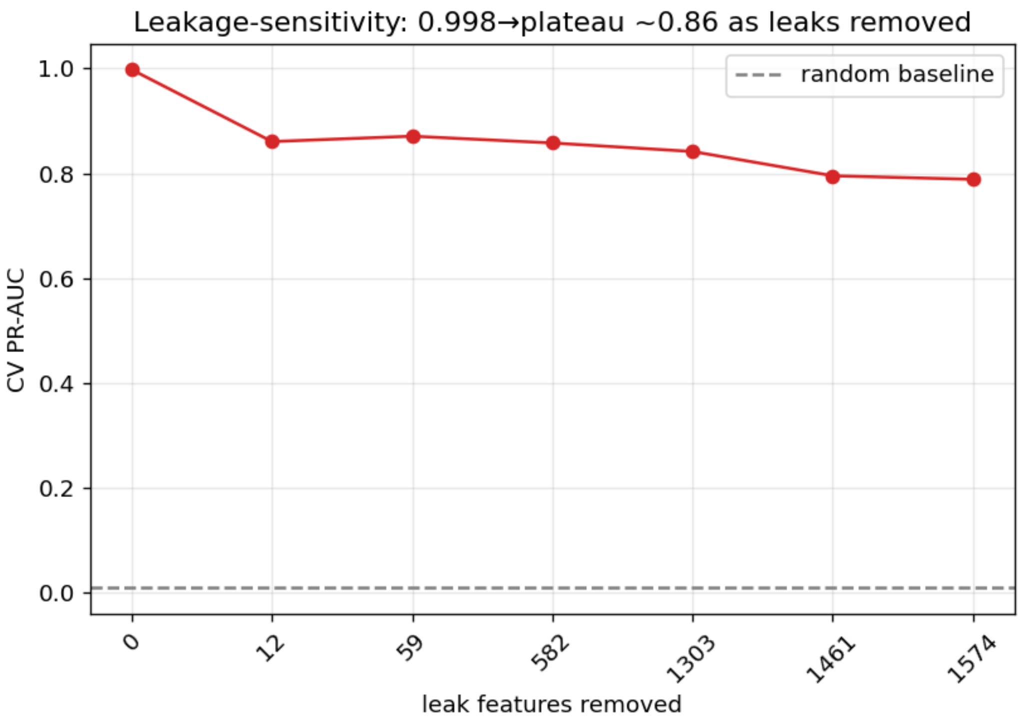

Fig 6. Leakage-sensitivity: score collapses 0.998 -> ~0.86 as leaks are removed, then PLATEAUS - proving the remaining signal is genuine, not one hidden leak.

6. See It Work - Live System Output

These are REAL outputs from our running engine for mule account #9003 (not mockups) - the exact score, SHAP reasons, investigation report and API response an analyst/system receives.

Analyst workflow (how it is used)

Open the dashboard -> accounts ranked by risk -> click a flagged account -> read its score, the plain-English SHAP reasons and peer-deviation -> trigger the recommended containment action (hold / escalate) with the auto-generated report attached as evidence. The screens below are the exact steps in that flow.

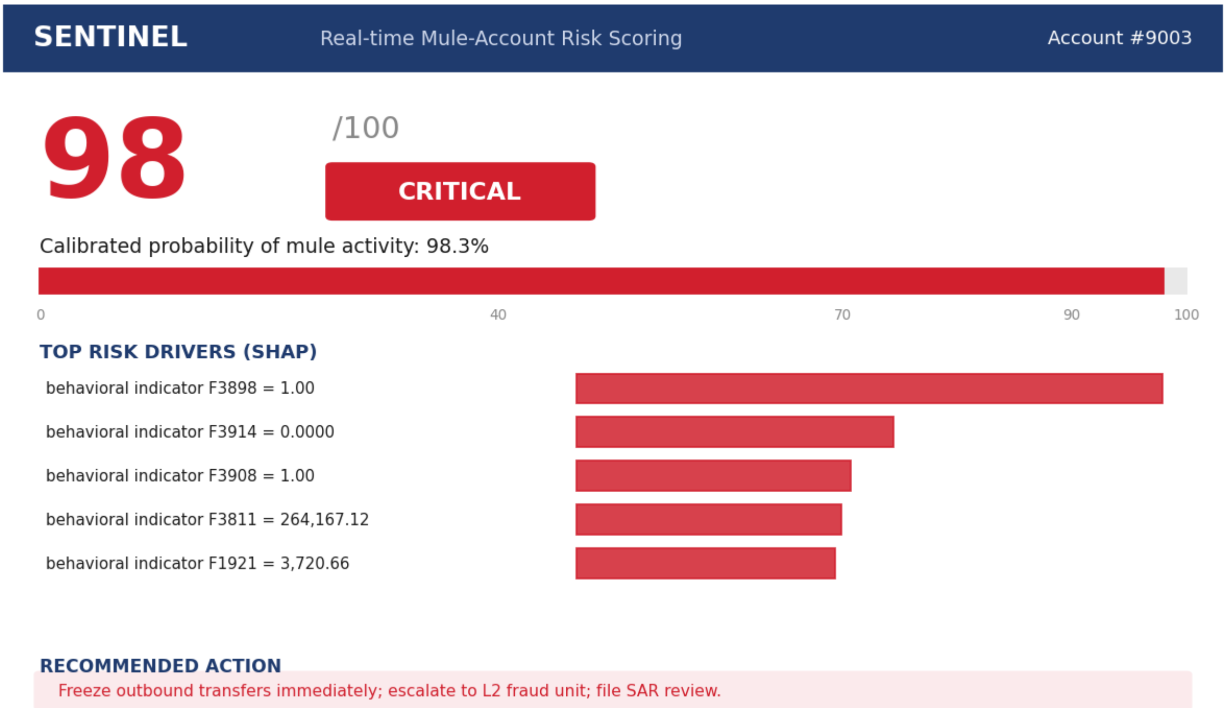

Fig 7. Real-time scoring card: a calibrated risk score, severity band, the SHAP drivers behind it, and a recommended action.

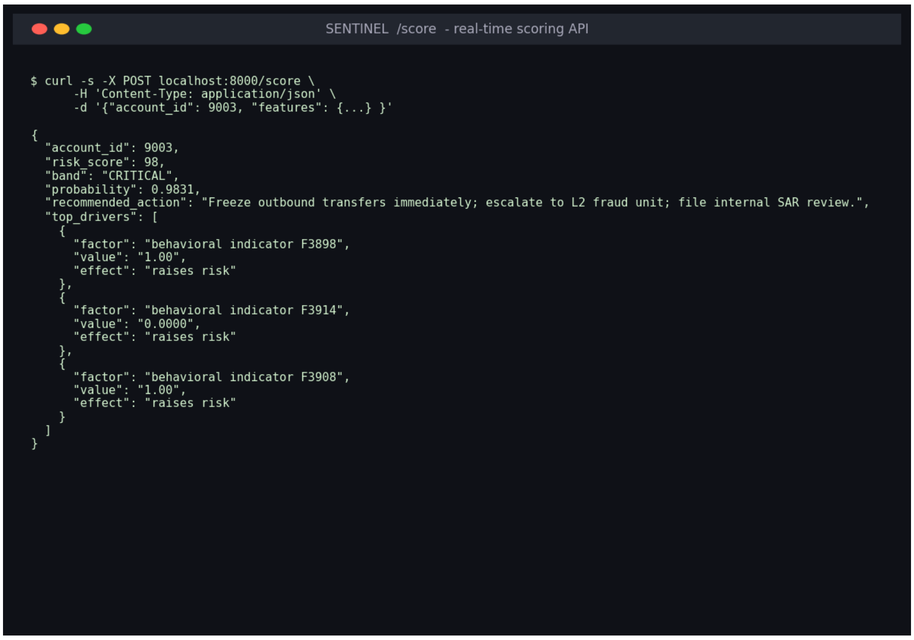

Fig 8. The live /score REST API response (FastAPI) - drop-in for any monitoring system.

6. See It Work (continued)

AUTO-GENERATED INVESTIGATION REPORT

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INVESTIGATION REPORT — Account #9003
========================================================
Risk Score : 98/100  (CRITICAL)
Calibrated probability of mule activity : 98.3%
Recommended action : Freeze outbound transfers immediately; escalate to L2 fraud unit; file
internal SAR review.

WHY THIS ACCOUNT WAS FLAGGED (top risk drivers):
  1. behavioral indicator F3898 (anonymized): 1.00 — below the median; closer to the typical
mule profile
  2. behavioral indicator F3914 (anonymized): 0.0000 — in the bottom 5% of accounts; closer to
the typical mule profile
  3. behavioral indicator F3908 (anonymized): 1.00 — in the top 5% of accounts; closer to the
typical mule profile
  4. behavioral indicator F3811 (anonymized): 264,167.12 — below the median; closer to the
typical mule profile
  5. behavioral indicator F1921 (anonymized): 3,720.66 — below the median; closer to the typical
mule profile

CONFIDENCE & CAVEATS:
  - Score derived from 2233 populated features; 732 were blank for this account.
  - Snapshot-based; not a substitute for transaction-level review.
  - F3912 (post-hoc fraud flag) was excluded as leakage — this score reflects behavioral signal
only.
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Fig 9. The one-click, auto-generated investigation report - analyst-ready, with confidence caveats and the leakage note built in.

7. Mule-Ring Detection - from accounts to networks (prototype)

Banks don't just want one flagged account - they want the RING. The provided data has NO explicit link data (shared device / IP / beneficiary / transactions), so we build a BEHAVIORAL-SIMILARITY graph: an edge joins two mules that are near-identical on the genuine (leak-removed) features - a data-grounded proxy for those links. Connected components = candidate rings to investigate together.

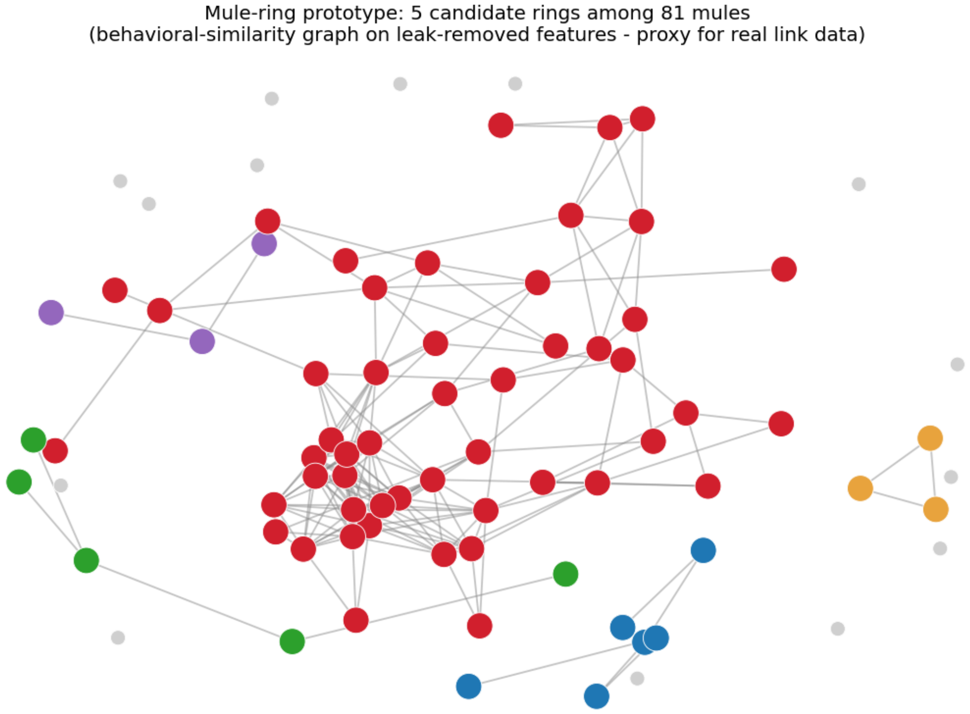

Fig 10. Working prototype on the provided data: 67 of 81 mules share a tight behavioral twin, forming 5 candidate rings - incl. one dominant 50-account cluster a desk would investigate as a batch.

Result (real, on this dataset): 67/81 mules cluster into 5 candidate rings; the largest is 50 behaviourally near-identical accounts. This is the SAME engine that, fed real shared-device / beneficiary / transaction edges (Phase-2), becomes production mule-NETWORK detection - we show the capability today rather than only promising it.

Is it a real ring? What we can and cannot claim (honest)

We CANNOT claim these are confirmed criminal rings - that needs transaction / device-link data we do not have. What we CAN show, measured on this data: (1) candidate Ring #1's members are ~30x more similar to each other than a random legit group (mean pairwise similarity 0.43 vs 0.01, well above the 95th-percentile chance ceiling of 0.04); (2) the cluster is STABLE - it recurs at 82% overlap (Jaccard) when 20% of features are randomly dropped, so it is not an artefact of one feature; (3) it is a PROXY for real links. We show the capability today; confirmation needs bank link data.

Case study: how SENTINEL works a candidate ring (real account from this data)

Account #9003 is flagged -> risk score 100 / CRITICAL (Section 6 shows its live scoring card & SHAP reasons). SENTINEL then asks the network question: #9003 is a central node in candidate Ring #1 - 50 behaviourally near-identical accounts. Potential exposure (estimate, configurable bank assumption): ~Rs 1.25 crore at Rs 2.5L average mule loss. Analyst action: don't chase one account - investigate the candidate ring as a batch, with the auto-built report as evidence. One alert -> a whole candidate ring surfaced for review.

8. Business Impact & Intelligent Alerts

Rupee-cost decision engine (banks act on money, not PR-AUC)

Translating the threshold into money (assumptions, configurable: avg mule loss Rs 2,50,000; analyst review Rs 400; false-positive harm Rs 5,000): at a practical operating point we catch 85-89% of mules for a net ~Rs 1.7 crore saved per 9,082-account population. Net savings is flat across thresholds, so the operating point is analyst-capacity-bound - the bank chooses the recall/alert-volume trade-off it can staff.

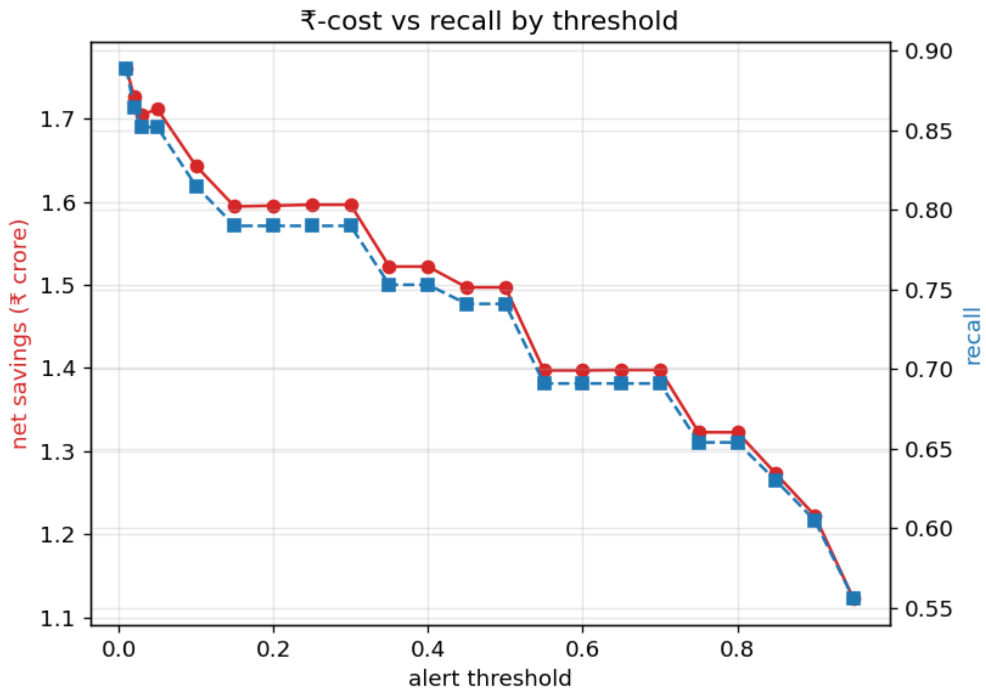

Fig 11. Net Rs-savings and recall vs alert threshold.

Analyst workload reduction (the value a fraud desk feels first)

Banks don't buy a model - they buy back analyst hours and caught fraud. SENTINEL turns an un-triageable book into a short, ranked, explained queue:

Stage	Accounts	What the analyst does
Whole book	9,082	impossible to review manually
True mules hidden inside	81 (0.89%)	the needles to find
SENTINEL top-50 watchlist	50	review THIS, ranked + explained
True mules in that watchlist	50 / 50	every reviewed account was a real mule (OOF)

Effect: the analyst reviews ~50 explained accounts instead of 9,082 - a >99% triage reduction on this data - and each comes with plain-English reasons and a recommended action, so a decision takes minutes. That operational lift (hours saved + fraud caught) is the primary purchase driver; the rupee and market figures below quantify it.

Intelligent, explainable alerts (PS2's 'intelligent alert generation')

Every alert carries: a 0-100 risk score + severity band (LOW/MEDIUM/HIGH/CRITICAL), the top SHAP-driven reasons in plain English, the account's deviation vs its peer cohort, a recommended action, and an auto-generated investigation report - so an analyst acts in minutes, not hours.

Go-to-market & sustainability (B2B / regulated)

Users & buyers: PSU & private banks' fraud, AML and transaction-monitoring desks (buyer: CRO / Head of

Fraud), plus payment bodies (NPCI) and regulator programmes (RBIH).

Deployment: on-prem / bank-VPC - data never leaves the bank - integrating with core banking and the existing transaction-monitoring + case-management systems; scoring on-demand via the REST API.

Adoption path: single-bank pilot on the bank's own book (prove the watchlist on real labels) -> production scoring -> multi-bank consortium / federated intelligence - the same distribution model RBIH uses for MuleHunter.AI.

Revenue & sustainability: per-bank annual licence + a per-account-scored usage tier; the shared cross-bank intelligence layer is regulator / consortium-funded. The cost to a bank is a small fraction of the fraud prevented (~Rs 1.7 cr per 9,082 accounts above). Acquisition is pulled by regulator credibility and a measurable ROI, with a low-friction pilot on the bank's existing data.

9. Market, Business Model & Traction

Market opportunity (illustrative, bottom-up)

Bottom-up estimates with explicit assumptions; exact market value to be finalised from RBI Annual Report fraud statistics, I4C / NCRP and industry AML-tech reports.

Tier	Definition	Size (illustrative)
TAM	All Indian banks + payment cos needing mule/AML scoring	~Rs 300-350 cr/yr
SAM	Scheduled commercial + payment + SFBs (~120, real-time-capable)	~Rs 120 cr/yr @ ~Rs 1 cr ACV
SOM	3-year capture: design partners -> ~20 banks	~Rs 18-20 cr ARR

Context: RBI-reported bank frauds run into tens of thousands of crores a year and mule accounts are the conduit - this is regulatory-mandated spend, not discretionary.

Business model & unit economics (illustrative)

Metric	Value*	Basis / assumption
ACV / bank / yr	Rs 0.8-1.2 cr	enterprise licence + per-account usage tier
Gross margin	~80%	software-led; cloud/compute + support
CAC / bank	~Rs 20-30 L	enterprise sales + pilot + integration
LTV (5-yr)	~Rs 3.5 cr	ACV x margin x retention
LTV : CAC	~12-15x	well above the 3x benchmark
CAC payback	< 6 months	post-landing

Year	Banks (cum.)	ARR*	Driver
Y1	3 (design partners)	~Rs 1.5 cr	discounted pilots; prove ROI on real books
Y2	10	~Rs 8 cr	regulator-backed referrals (RBIH / NPCI)
Y3	22	~Rs 18-20 cr	+ consortium shared-intelligence revenue

**Illustrative & assumption-driven - to be calibrated during paid pilots; not a forecast.*

Traction: technical validation today, customer validation next (honest)

WHAT WE HAVE (technical traction): a working end-to-end prototype, CI smoke tests (5/5), a deployable FastAPI service, a live Streamlit dashboard, reproducible results and a ranked watchlist (top-50 watchlist caught all 50 true mules). WHAT WE DO NOT YET HAVE: customers, paid pilots or LOIs - all traction so far is technical, and we say so plainly.

- Gate 1 (0-3 mo): 1-2 design-partner banks; run SENTINEL on their historical book; validate precision@50 on real labels; target 2-3 letters of intent.
- Gate 2 (3-9 mo): RBIH / NPCI engagement + regulatory sandbox; first paid pilot.
- Gate 3 (9-18 mo): production deployment at 1-2 banks; published case study; consortium pilot.

Why this is winnable, not wishful: a hard regulatory mandate (RBI MuleHunter.AI), a low-friction pilot that runs on the bank's own data on-prem, and a measurable rupee ROI - the three levers that shorten enterprise-bank sales cycles.

10. Feasibility, Prototype Roadmap & Honesty

Already working (reproducible)

- End-to-end pipeline (preprocess -> auditor -> model -> finalize -> insights -> scoring).
- Real-time FastAPI /score + /report service (~35 ms) and a Streamlit analyst dashboard.
- Deliverables: scored predictions for all 9,082 accounts + a ranked watchlist with reasons.
- Model card, smoke tests (5/5 passing), pinned requirements, and a self-contained Colab notebook.

Reproducible end-to-end - run it yourself.

Repo: https://github.com/vibhukrishnas/sentinel-mule-detection

Live demo: https://sentinel-mule-detection-hvymcown8cyerjjes48cka.streamlit.app

Phase-2 prototype plan (1 Jul - 17 Aug)

- GRAPH mule-RING detection: PROTOTYPE already built (behavioral-similarity graph, see Section 7); production version ingests real shared device / IP / beneficiary + circular-flow edges.
- Multi-feed fusion: live transaction streams + fraud / txn-monitoring alerts + govt cyber-fraud tickets + RBI/regulatory feeds, scored in real time to PREVENT CIRCULATION.
- Confirm with the bank which near-label features are pre- vs post-event (locks the exact number); plug in real cost figures for a per-channel cost-optimal threshold.
- Federated, privacy-preserving cross-bank intelligence; drift monitoring + scheduled leak re-audit.
- Hardening: streaming + batch scoring, alert-workflow / case-management integration, audit logging.

What we explicitly do NOT claim (maturity signal)

- Not a transaction stream (snapshot data) - we say so plainly.
- Anonymised features -> semantic labels in narratives are inferred, not bank-confirmed.
- 81 positives -> estimates carry real variance; we report a range and a confidence interval.
- Decision-support for analysts, not an autonomous freeze authority.

Anticipated questions (and our answers)

- Why is the model so good? We removed target leakage (F3912 + ~585 leaks via the auditor); the honest score is 0.81-0.89, not 1.0.
- Built today vs future? SOLID boxes in the architecture = built on the provided data; DASHED = Phase-2 (live feeds + graph mule-rings).
- Deployment realism? Decision-support that recommends actions to an analyst - NOT an autonomous freeze authority.
- Live fraud data? The prototype is trained on the PROVIDED SNAPSHOT; live transaction / alert / regulatory feeds are Phase-2.
- Why not deep learning? 81 positives, ~3,900 sparse features, 27.6% NaN, and the need for fast, interpretable scoring make LightGBM the right fit; DL would overfit.

Bottom line: a fake 1.000 is easy on this data. We present a defensible 0.81-0.89, a top-50 watchlist that caught all 50 true mules, candidate mule-rings, a rupee impact, and the leakage audit that proves we earned it.